



FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF BUSINESS INFORMATION TECHNOLOGY
END OF SEMESTER EXAMINATION
BBT 2104: INTEGRAL CALCULUS

Date: 19 July, 2017

Time: 2 Hours

Instructions

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

QUESTION ONE (30 MARKS)

- (a) Use the Midpoint Rule with $n = 5$ to approximate $\int_0^1 (x^4 + 1) dx$ [3 Marks]
- (b) Evaluate the following integrals:
- (i) $\int x(2x^2 + 3)^4 dx$ [3 Marks]
- (ii) $\int \frac{2x^6}{x^7 + 3} dx$ [3 Marks]
- (iii) $\int_0^{\pi} \cos^2 x \sin x dx$ [3 Marks]
- (iv) $\int_0^1 x e^x dx$ [3 Marks]
- (c) Find the area of the region lying above the x-axis and under the parabola $y = 4x - x^2$. [5 Marks]
- (d) Find a power series expansion about $\pi/2$ for $f(x) = \cos x$ [5 Marks]
- (e) Using trigonometric substitution, $x = 2 \tan \theta$, evaluate

$$\int \frac{dx}{x^2 \sqrt{4 + x^2}}$$

[Hint: $\sec^2 x = 1 + \tan^2 x$]

[5 Marks]

QUESTION TWO (20 MARKS)

- (a) Solve the following initial value problem.

$$\begin{cases} \frac{dy}{dx} = y^2 x \\ y(0) = 1 \end{cases}$$

[5 Marks]

- (b) Consider the solid of revolution obtained by revolving about the x-axis the region in the first quadrant bounded by the parabola $y^2 = 8x$ and the lines $x = 2$. Using the disk formula, find the volume of the resultant solid of revolution

[5 Marks]

- (c) Use an appropriate method to evaluate the following definite integrals:

(i) $\int_0^4 \sqrt{2x+1} \, dx$

[5 Marks]

(ii) $\int_0^1 \sin^{-1} x \, dx$

[5 Marks]

QUESTION THREE (20 MARKS)

- (a) If 20% of a radioactive substance disappears in one year, find the half-life T .

Assume exponential decay.

[4 Marks]

- (b) Solve the following differential equation:

$$\frac{dy}{dx} = \frac{2x^2 + 4x + 1}{x + 2}(y - 3),$$

given that $y(0) = 7$.

[6 Marks]

- (c) Use integration by parts to verify the specified formula.

$$\int e^x \sin x \, dx = \frac{1}{2}e^x(\sin x - \cos x) + C$$

[6 Marks]

- (d) Find: $\int (2x + \sin^2 x) \, dx$

[4 Marks]

QUESTION FOUR (20 MARKS)

- (a) Determine the area of the region bounded by the given curves $y = 9 - x^2$ and the line $x + 3 = y$.

[7 Marks]

- (a) If the population of a country is 50 million people, and the population is increasing exponentially with a growth constant $k = \ln 2$, calculate precisely the population after 5 years.

[3 Marks]

(c) Decompose the following rational function into partial fractions:

$$R(x) = \frac{x^3 + x^2 + 2}{x^2 - 1},$$

[3 Marks]

and hence evaluate,

$$\int \frac{x^3 + x^2 + 2}{x^2 - 1} dx$$

[7 Marks]

QUESTION FIVE (20 MARKS)

(a) Find the area of the region between $y = x(x^2 - 6x + 8)$ and the x-axis.

[7 Marks]

(b) Resolve the following rational into partial fractions

$$R(x) = \frac{x^3 - 6x + 3}{(x - 2)^3}.$$

[5 Marks]

(c) Use an appropriate method to evaluate the following integrals:

(i) $\int \cot x \, dx$ [Hint: $\cot x = \frac{1}{\tan x}$]

[3 Marks]

(ii) $\int \sqrt{1 - x^2} \, dx$ [Hint: Let $x = \sin(\theta)$ and $\cos^2 x = \frac{1}{2} + \frac{\cos 2x}{2}$]

[5 Marks]

END OF PAPER